

Sehar Tariq

AI Student | Machine Learning & Applied Research Enthusiast

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Undergraduate AI student building applied ML and NLP systems, including a RAG-based healthcare chatbot and a real-time activity/vital-signs classifier. Co-authored a survey study on blended learning (74% approval rate). Experienced in research methodology and data analysis. Seeking internships and entry-level roles in AI/ML, data science, and research.

SKILLS

- Python (Pandas, NumPy, Scikit-learn)
- Machine Learning & Deep Learning (TensorFlow, Keras)
- Data Analysis & Visualization (SPSS, R, Matplotlib, Seaborn)
- SQL & Database Design
- Natural Language Processing (RAG)
- Git & GitHub
- Research Methodology & Scientific Writing
- Image Processing (OpenCV)
- Team collaboration & volunteering
- Microsoft Office (Excel, Word, Powerpoint)

PROJECTS

Oncology RAG-based Chatbot | Islamabad

Sep, 2025

Built a medical information system using Retrieval-Augmented Generation for efficient healthcare data retrieval.

- Designed and implemented a multi-modal data pipeline integrating OCR (PyTesseract) and speech-to-text for document ingestion
- Integrated FAISS vector database and Groq API for fast and accurate information retrieval
- Developed a responsive user interface using Gradio, enabling real-time interaction with the chatbot
- Presented working prototype to faculty and peers, demonstrating end-to-end system functionality

Intelligent Activity Recognition & Physiological Estimation | Islamabad

Jan, 2026

Developed a real-time activity classifier and vital signs estimator using PPG signals and deep learning.

- Built an MLP-based deep learning model achieving 75% accuracy in activity classification
- Implemented heart rate and SpO2 estimation from PPG signal data using neural networks
- Applied SHAP (Explainable AI) for model interpretability and feature importance analysis
- Created an interactive Streamlit dashboard for real-time visualization of predictions

Blended Learning Research Paper (Co-Author) | Islamabad

Dec, 2024

Conducted a survey-based study on student perceptions of blended learning.

- Designed and distributed surveys to university students; analyzed responses using Python and statistical methods
- Identified a 74% approval rate for flexible learning and key challenges in technical support
- Co-authored the full research paper, presenting findings in clear, non-technical language
- Gained hands-on experience in research design, data collection, and academic writing

PROFESSIONAL EXPERIENCE .

Special Communications Organization (SCO) AI-Intern	Rawalakot, AJK <i>June, 2025 – Sep, 2025</i>
- Developed a domain-specific RAG chatbot for oncology information retrieval, later carried forward as a full prototype (see Projects) - Built and tested the data ingestion and retrieval pipeline under senior guidance in a professional setting	

EDUCATION .

Air University Islamabad, Pakistan BS Artificial Intelligence	<i>Sep, 2023 – June, 2027 (expected)</i>
Govt. Post Graduate College Rawalakot, AJK F.Sc. (Pre-Medical)	<i>Sep, 2019 – June, 2022</i>

COURSES .

Systematic Review and Meta-Analysis – Multiomics Oct 2025 Scientific Writing Internship Program (Cohort 6) – Multiomics Aug–Nov 2025 Data Analysis Through SPSS & R – Multiomics Aug–Sep 2025	
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ADDITIONAL SKILLS & ACTIVITIES .

Volunteering: Usher at Indus AI Week 2026, Pakistan; Decor Team Member at Air University, Islamabad (collaborated on campus event setup and management, contributing to university-wide events) Event: Workshop: Vibecoding with Google Antigravity, GDGOC DevSummit 2026, Bahria University Islamabad Languages: English (Fluent), Urdu (Native) Soft Skills: Problem-Solving, Team Collaboration, Communication, Adaptability	
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